

Mapping the 2D Classical Ising Model to the 1+1D Quantum Ising Model

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1 From Classical Partition Functions to Quantum Circuits

This note is devoted to the explicit mapping between the 2D classical Ising model and the 1+1D quantum Ising model. This is a classic example of the deep connection between classical statistical mechanics and quantum mechanics, as the slogan say

$$D + 1 \text{ dim space statistical mechanics} = D \text{ dim space quantum mechanics} \quad (1)$$

which can be understood through the **transfer matrix** method and **tensor networks**. The key idea is that the partition function of a classical model can be expressed as a product of transfer matrices, which in turn can be interpreted as layers of a quantum circuit. By decomposing the transfer matrix into simpler building blocks, we can identify the corresponding quantum gate. Below we will illustrate this mapping concretely for the 2D classical Ising model on a square lattice, and show how the quantum Hamiltonian emerges.

1.1 The transfer matrix and quantum circuits

A central idea connecting classical statistical mechanics to quantum mechanics is the *transfer matrix* method. Consider a classical statistical model defined on a lattice. Its partition function can be written as

$$Z = \text{tr}(e^{-\beta H}) = \text{tr}(T^N), \quad (2)$$

where T is the transfer matrix and N is the number of rows (or time slices). The key observation is that the transfer matrix of a classical model plays the same role as the time-evolution operator $e^{-\tau \hat{H}}$ in quantum mechanics (in imaginary time). In other words, each application of T corresponds to one step of imaginary-time evolution – or equivalently, one layer of a quantum circuit.

This transfer-matrix/quantum-circuit correspondence provides a powerful dictionary:

Classical statistical system	Quantum system
Partition function $Z = \text{tr}(T^N)$	Imaginary-time path integral
Transfer matrix T	Imaginary-time evolution $e^{-\tau \hat{H}}$
Row-to-row transfer	One layer of quantum circuit
Classical spins $\sigma_i = \pm 1$	Computational basis $ \sigma_i\rangle$
Inverse temperature β	Imaginary time τ

We now illustrate this correspondence concretely for the 2D classical Ising model on a square lattice.

1.2 The 2D classical Ising model

Consider a square lattice with classical Ising spins $\sigma_i = \pm 1$ placed on each vertex. The Hamiltonian is

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \quad (3)$$

where the sum is over nearest-neighbor pairs. We split the interactions into *horizontal* bonds (within a row) and *vertical* bonds (between adjacent rows). Denoting the spins in row n collectively as $\sigma_n = (\sigma_{n,1}, \sigma_{n,2}, \dots, \sigma_{n,L})$, the partition function on an $N \times L$ lattice with periodic boundary conditions is

$$Z = \sum_{\{\sigma\}} \prod_{n=1}^N \exp \left(\beta_h \sum_j \sigma_{n,j} \sigma_{n,j+1} + \beta_v \sum_j \sigma_{n,j} \sigma_{n+1,j} \right), \quad (4)$$

where $\beta_h = \beta J_h$ and $\beta_v = \beta J_v$ are the horizontal and vertical couplings respectively. In the isotropic case $\beta_h = \beta_v \equiv K$.

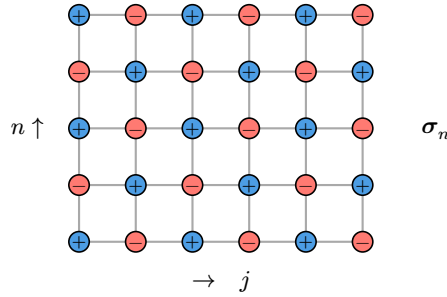


Figure 1: An configuration (paramagnetic phase) of classical Ising spins $\sigma_i = \pm 1$ on each vertex on square lattice. Blue (+) and red (−) denote spin-up and spin-down respectively. Horizontal bonds carry coupling β_h and vertical bonds carry coupling β_v , where n takes value $1, 2, \dots, N$, and j takes value $1, 2, \dots, L$.

1.3 Decomposing the transfer matrix into tensor building blocks

The transfer matrix T maps one row of spins to the next. Its matrix elements are

$$\langle \sigma' | T | \sigma \rangle = \exp \left(\beta_h \sum_j \sigma'_j \sigma'_{j+1} + \beta_v \sum_j \sigma_j \sigma'_j \right). \quad (5)$$

The crucial insight is that we can factorize T into a product of two simpler pieces as in figure 2:

$$T = T_{\text{vertical}} \cdot T_{\text{horizontal}} = \prod_j e^{\beta_v \sigma_j \sigma'_j} \cdot \prod_j e^{\beta_h \sigma'_j \sigma'_{j+1}}. \quad (6)$$

Now we promote the classical spins to quantum operators. In the computational basis $|\sigma\rangle$ (eigenstates of $\hat{Z}$), we identify:

- **Diagonal interaction (vertical bond):** The Boltzmann weight $e^{\beta_v \sigma_j \sigma'_{j+1}}$ acts *between* two adjacent rows. When interpreted as a matrix in the σ, σ' basis, it takes the form:

$$T_{\text{vert}} = \begin{pmatrix} e^{\beta_v} & e^{-\beta_v} \\ e^{-\beta_v} & e^{\beta_v} \end{pmatrix} = e^{\beta_v} (1 + e^{-2\beta_v} X) \propto e^{\beta_v^* \hat{X}}, \quad (7)$$

where $\hat{X}$ is the Pauli- X operator and the dual coupling β_v^* is defined by

$$e^{-2\beta_v^*} = \tanh(\beta_v), \quad \text{i.e.,} \quad \beta_v = \text{arctanh}(e^{-2\beta_v^*}). \quad (8)$$

To verify: $e^{\beta_v^* \hat{X}} = \cosh(\beta_v^*) \hat{I} + \sinh(\beta_v^*) \hat{X}$, which has matrix elements $e^{\beta_v^* \hat{X}}_{\sigma\sigma'} \propto e^{\beta_v \sigma \sigma'}$.

- **Off-diagonal interaction (horizontal bond):** The Boltzmann weight $e^{\beta_h \sigma_j \sigma_{j+1}}$ acts *within* a single row. Since σ_j and σ_{j+1} are in the $\hat{Z}$ -eigenbasis, this is simply

$$T_{\text{horiz}} = e^{\beta_h \hat{Z}_j \hat{Z}_{j+1}}. \quad (9)$$

Therefore the full transfer matrix is a product of commuting two-body gates:

$$T = \prod_j e^{\beta_v^* \hat{X}_j} \cdot \prod_j e^{\beta_h \hat{Z}_j \hat{Z}_{j+1}}, \quad (10)$$

which is *exactly* one layer of a quantum circuit composed of single-qubit rotations $e^{\beta_v^* X}$ and nearest-neighbor ZZ -interactions $e^{\beta_h ZZ}$.

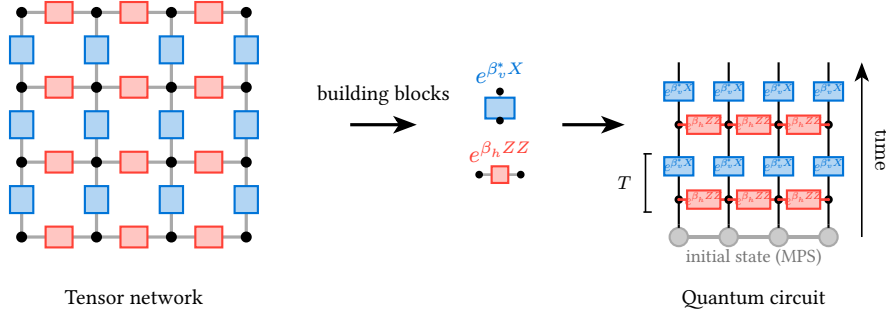


Figure 2: Left: tensor network representation of the partition function, with blue tensors ($e^{\beta_v^* X}$) on vertical bonds and red tensors ($e^{\beta_h ZZ}$) on horizontal bonds. Middle: the two building blocks. Right: the corresponding quantum circuit, where each layer represents one application of the transfer matrix.

1.4 The quantum Hamiltonian

In the limit where the vertical coupling is much stronger than the horizontal one (the *anisotropic* limit), we can write $T \approx e^{-\tau \hat{H}}$ with a small imaginary-time step τ (revert trotterization), so that

$$\hat{H} = -J \sum_j \hat{Z}_j \hat{Z}_{j+1} - h \sum_j \hat{X}_j, \quad (11)$$

which is exactly the 1D transverse-field quantum Ising model. The couplings are related by

$$\beta_h \rightarrow J\tau, \quad \beta_v^* \rightarrow h\tau. \quad (12)$$

This establishes the complete mapping:

2D classical Ising model ($N \times L$ square lattice, coupling constant K)

$\Downarrow$

1+1D quantum Ising model (L sites, N imaginary-time steps, coupling constant h)

$$\hat{H} = -J \sum_j \hat{Z}_j \hat{Z}_{j+1} - h \sum_j \hat{X}_j \quad (13)$$

Physical observables are mapped as follows:

- **Free energy** Taking the partition function $Z = \text{tr}(T^N)$, the free energy is $F = -\frac{1}{\beta} \ln Z = -\frac{1}{\beta} N \ln \lambda_0$. Its density corresponds to the ground-state energy E_0 :

$$f = -\frac{1}{\beta} \frac{\ln Z}{N} = -\frac{1}{\beta} \ln T \propto E_0 = -\frac{1}{\beta} \ln \lambda_0, \quad (14)$$

where λ_0 is the largest eigenvalue of T , and the ground-state energy E_0 of the quantum Hamiltonian is related to λ_0 .

- **Correlation function** $\langle \sigma_i \sigma_j \rangle$ in the classical model (note here we need to insert two spin operators among space directions) corresponds to $\langle \hat{Z}_i \hat{Z}_j \rangle$ in the quantum model.
- **Phase diagram:** On the microscopically isotropic line $\beta_h = \beta_v = K$, the 2D classical Ising model has a ferromagnetic (FM) to paramagnetic (PM) phase transition at $K_c = \frac{1}{2} \log(1 + \sqrt{2})$. In the Hamiltonian (strongly anisotropic) limit, the classical criticality condition instead becomes $(h/J)_c = 1$ for the transverse-field Ising chain. These are two different parameterizations of the same Ising universality class, with central charge $c = 1/2$.

1.5 Self-duality, spacetime isotropy, and criticality

Three notions must be kept separate. We restrict here to the uniform, zero-field, ferromagnetic square-lattice Ising model, $\beta_h, \beta_v > 0$, and take the horizontal and vertical lattice spacings to be equal.

1. **Self-dual locus (a microscopic duality condition).** Define the dual of a coupling K by

$$e^{-2K^*} = \tanh K, \quad \text{equivalently} \quad \sinh(2K) \sinh(2K^*) = 1. \quad (15)$$

Kramers-Wannier duality exchanges horizontal bonds with dual vertical bonds and vice versa:

$$\sinh(2\beta_h) \sinh(2\tilde{\beta}_v) = 1, \quad \sinh(2\beta_v) \sinh(2\tilde{\beta}_h) = 1. \quad (16)$$

Thus $(\tilde{\beta}_h, \tilde{\beta}_v) = (\beta_v^*, \beta_h^*)$. Requiring the dual model to have the same oriented couplings gives

$$\sinh(2\beta_h) \sinh(2\beta_v) = 1. \quad (17)$$

This statement concerns invariance of the microscopic Boltzmann weights under duality; by definition it says nothing about a divergent correlation length.

2. **Microscopic spacetime-isotropic locus (a rotation condition).** Exchanging the two lattice directions is a symmetry when

$$\beta_h = \beta_v. \quad (18)$$

This is the diagonal of the coupling plane. It is not a phase boundary: varying K along $(\beta_h, \beta_v) = (K, K)$ passes from the PM phase through one critical point and then into the FM phase. Notice especially that $\beta_h = \beta_v$ is not the same equation as $\beta_h = \beta_v^*$. The latter is equivalent to the self-dual equation above.

3. **Critical locus (an infrared condition).** Criticality means that the bulk correlation length diverges, $\xi \rightarrow \infty$, or equivalently that the transfer-matrix gap closes in the thermodynamic limit. Kramers-Wannier duality maps the critical locus to itself, but duality alone does not in general prove that every self-dual point is critical. For this particular clean Ising model, the exact solution (together with the uniqueness of the FM–PM transition in the positive-coupling quadrant) gives

$$\mathcal{C}_{\text{crit}} = \{(\beta_h, \beta_v) : \sinh(2\beta_h) \sinh(2\beta_v) = 1\}. \quad (19)$$

Hence the self-dual and critical loci coincide *as subsets of this model's coupling plane*, although their definitions and physical content are different.

The microscopic spacetime-isotropic locus intersects the critical locus only once. Setting $\beta_h = \beta_v = K$ gives

$$\sinh^2(2K_c) = 1 \quad \Rightarrow \quad K_c = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407. \quad (20)$$

There is a second, continuum-level use of spacetime isotropy.’’ Every point on the anisotropic critical curve has the Ising CFT as its infrared fixed point. Its bare anisotropy appears as a nonuniversal velocity, or equivalently as a background metric, and can be removed by rescaling one coordinate. This emergent isotropy after coordinate rescaling does not imply the microscopic equality $\beta_h = \beta_v$.

Finally, in the Hamiltonian limit used above,

$$\beta_h = \tau J + O(\tau^2), \quad \beta_v^* = \tau h + O(\tau^2), \quad (21)$$

so the self-dual/critical condition becomes $\beta_h = \beta_v^*$, and therefore $J = h$ as $\tau \rightarrow 0$. This is the quantum critical condition. It must not be confused with the microscopic spacetime-isotropic condition $\beta_h = \beta_v$: the Hamiltonian limit instead has $\beta_h \rightarrow 0$ and $\beta_v \rightarrow \infty$.

Concept	Definition	Physical meaning
Self-dual locus	$\sinh(2\beta_h) \sinh(2\beta_v) = 1$	Fixed locus of Kramers-Wannier duality; a microscopic algebraic condition.
Microscopic spacetime-isotropic locus	$\beta_h = \beta_v$	Exchange symmetry of the two lattice directions; not a phase boundary.
Critical locus	$\xi \rightarrow \infty$	Infrared condition. For this model it coincides with the self-dual locus and flows to Ising CFT.
Isotropic critical point	$(\beta_h, \beta_v) = (K_c, K_c)$	Unique intersection of the microscopic isotropic and critical loci.

The full phase diagram has:

- **Ferromagnetic (FM) phase:** $\sinh(2\beta_h) \sinh(2\beta_v) > 1$; the $\mathbb{Z}_2$ spin-flip symmetry is spontaneously broken in the thermodynamic limit.
- **Paramagnetic (PM) phase:** $\sinh(2\beta_h) \sinh(2\beta_v) < 1$; the $\mathbb{Z}_2$ symmetry is unbroken.
- **Critical locus:** $\sinh(2\beta_h) \sinh(2\beta_v) = 1$. Every point belongs to the Ising CFT universality class with $c = 1/2$, but only (K_c, K_c) is microscopically invariant under exchanging the two spacetime lattice directions.

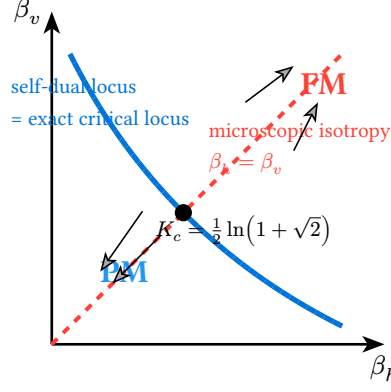


Figure 3: Schematic phase diagram of the zero-field ferromagnetic square-lattice Ising model in the β_h – β_v plane. Blue denotes the self-dual locus. The exact solution shows that, for this model, the same set is also the critical locus separating the FM and PM phases; this equality is a result, not a definition. Red denotes the microscopic spacetime-isotropic locus $\beta_h = \beta_v$, which crosses the critical locus only at (K_c, K_c) , where $K_c = \frac{1}{2} \ln(1 + \sqrt{2})$. All points on the critical locus have the Ising CFT as their infrared theory after a nonuniversal coordinate rescaling. Gray arrows indicate schematic RG flows.

1.6 Remarks

- The transfer matrix approach generalizes to other classical models. For example, the q -state Potts model maps to quantum chains with $SU(q)$ symmetry, and the classical XY model maps to the quantum rotor model.
- The cylindrical geometry (periodic boundary conditions in the spatial direction) of the classical model maps directly to a quantum system on a ring, where the transfer matrix wraps around to form a cylinder in the tensor-network picture. Such geometry is particularly natural for extracting the central charge c from finite-size scaling of the free energy.